Project Idea Description

Smart Digital Receipt and Food Expiration Management Application

In modern daily life, people frequently purchase groceries from supermarkets and retail stores. After completing a purchase, customers usually receive a **paper receipt** that records information such as the purchased items, prices, purchase date, and store location. However, these paper receipts are often **lost, discarded, or difficult to organize**, making it hard for consumers to track their historical spending or review what they have previously purchased.

This project proposes the development of a **Smart Digital Receipt and Food Expiration Management Application** that automatically records customers' purchase information and provides intelligent tools for spending analysis and food management.

The main goal of this system is to **digitize grocery receipts and provide a convenient way for users to track their purchases and manage food consumption** through a mobile application.

Core Concept of the System

The proposed system allows users to **automatically store grocery purchase information in a mobile application** when they shop in supermarkets.

During the checkout process, customers normally scan each product's barcode at the **Point-of-Sale (POS) machine** before completing payment. In the proposed system, users will also have a **digital membership card inside the mobile application.** When the customer checks out, they can scan this membership code at the POS machine.

After the purchase is completed, the POS system will automatically send the **digital receipt information** to the user's account in the mobile application.

The digital receipt will contain information such as:

- Purchase date and time
- Store location
- List of purchased products
- Product prices
- Total spending amount

This information will be automatically stored in the application database and displayed to the user in a structured format.

As a result, users no longer need to keep physical receipts or manually record their expenses.

Function 1: Automatic Expense Tracking

One important feature of the system is **automatic spending tracking**.

Because all purchases are recorded digitally, the application can analyze the user's spending behaviour over time.

The system will categorize purchased products into different spending categories, such as:

- Fruits and vegetables
- Meat and seafood
- Dairy products
- Snacks
- Beverages
- Household supplies

Based on this categorized data, the application can generate **spending summaries and visual reports** that allow users to better understand their consumption patterns.

For example, the application may display:

- Monthly grocery spending
- Spending by product category
- Purchase frequency for certain items
- Shopping locations visited most frequently

This function helps users **manage their personal finances more effectively** and make better purchasing decisions.

Function 2: Purchase History and Product Records

Another important function of the application is to provide a **complete purchase history** for the user.

The system will store all past receipts and allow users to review them at any time.

Users will be able to see:

- What they bought
- When they bought it
- Where they bought it
- How much they spent

This feature provides a convenient way to **review shopping habits, manage budgets, and verify past purchases**.

For example, if a user wants to check when they last purchased milk or how often they buy certain products, they can easily search their purchase history within the application.

Function 3: Food Expiration Reminder

Another key function of this system is to help users **manage food expiration dates** and reduce food waste.

Many grocery products, especially fresh food items, contain expiration dates printed on the packaging. However, consumers often forget about these dates after storing the food at home. As a result, food may expire before it is consumed.

In this system, when a product is purchased and recorded in the digital receipt, the application will also attempt to obtain the **expiration date information for that product**.

Once the expiration date is known, the system will store this information in the user's food inventory list.

The application will then monitor the expiration dates of purchased items and send **notifications or reminders** to the user when a product is close to expiring.

For example, the application may send reminders such as:

- "Your milk will expire in two days."
- "The vegetables you purchased last week may expire soon."

These reminders encourage users to **consume food before it expires**, which helps reduce unnecessary food waste.

Function 4: AI-Based Shopping Strategy Assistant

Another advanced feature of the proposed system is an **AI-based shopping strategy assistant** that helps users make better purchasing decisions in future shopping trips.

Since the application continuously collects and stores the user's grocery purchase history, the system can analyze this historical data to understand the user's **shopping habits, product preferences, and consumption patterns**. By applying data analysis and machine learning techniques, the application can generate intelligent suggestions that help users plan their future grocery purchases more effectively.

For example, the system can analyze information such as:

- Frequently purchased products
- Average purchase intervals for certain items
- Typical quantities purchased
- Spending patterns by category
- Food items that are often unused or expired

Based on this analysis, the AI assistant can provide **personalized recommendations** before or during the user's next shopping trip.

Some examples of possible recommendations include:

1. Smart Shopping List Suggestions
The system can automatically generate a suggested shopping list based on the user's previous purchasing behavior and estimated consumption rate. For example, if the user usually buys milk every seven days, the application may suggest adding milk to the shopping list when it predicts that the current supply is running low.

2. Avoiding Over-Purchasing
The application can warn users if they are about to purchase items that they already have at home. For example, if a user recently bought a large quantity of yogurt that has not yet been consumed, the system may display a reminder such as:
"Based on your previous purchases, you may still have yogurt at home."

3. Expiration-Aware Shopping Suggestions
The AI assistant can also consider food expiration information stored in the system. If some items at home are close to their expiration date, the system may recommend **not purchasing similar items again** until the existing ones are consumed.

4. Spending Optimization Suggestions
By analyzing past spending behavior, the system can help users adjust their shopping strategy. For example, if the user frequently spends a large portion of their grocery budget on snacks or beverages, the application may provide insights such as:
"You spent 25% of your grocery budget on snacks last month. Consider adjusting your purchase plan to manage your spending more effectively."

5. Personalized Shopping Insights
The AI assistant may also provide general insights about the user's shopping habits, such as identifying frequently purchased items, typical shopping frequency, and potential opportunities to reduce waste or save money.

Through these personalized suggestions, the AI assistant helps users **develop more efficient and sustainable grocery shopping strategies**, improving both financial management and household food consumption planning.

Benefits of the Proposed System

This system provides several benefits for consumers:

1. Paperless Receipts

Users no longer need to keep or organize physical receipts because all receipts are stored digitally.

2. Automatic Expense Tracking

The application records spending automatically without requiring manual input from the user.

3. Better Financial Awareness

Users can analyze their grocery spending patterns and manage their budgets more effectively.

4. Food Waste Reduction

By reminding users of upcoming expiration dates, the system helps people consume food before it spoils.

5. Convenient Purchase Records

Users can easily review past purchases and search for specific products.

Summary of the Idea

In summary, this project proposes a **mobile application that integrates digital receipts, expense tracking, and food expiration management into a single platform**.

By connecting the grocery checkout process with a mobile application through a digital membership system, users can automatically record their purchases and receive intelligent reminders related to their food consumption and spending habits.

The proposed system aims to improve **shopping convenience, personal financial management, and household food management** through the use of digital technology.

Technical Challenges

Although the concept of a smart digital receipt and food management system is promising, there are several technical challenges that need to be considered when implementing this idea in practice.

1. Limited Access to Point-of-Sale (POS) Systems

One of the initial ideas of this project is to allow the application to automatically receive digital receipt information from supermarket POS systems when users scan their membership code during checkout. However, in practice, accessing POS systems is difficult.

Most supermarket POS systems are **proprietary systems controlled by retailers**, such as supermarkets or payment providers. These systems are typically integrated with internal inventory management and payment infrastructure. As a result, external developers or third-party applications generally **do not have direct access to POS data or transaction records.**

Furthermore, implementing such integration would require cooperation with retailers and access to their internal APIs or data infrastructure, which is not feasible within the scope of a university student project. Therefore, relying on POS integration as the primary method for collecting receipt information presents a significant technical limitation.

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2. Limitations of Product Barcodes

Another challenge relates to the use of **product barcodes** for retrieving detailed product information such as expiration dates.

Most retail products use standard barcode formats such as **UPC or EAN codes**, which typically encode only a **product identifier (GTIN)**. This identifier allows the system to recognize the product type (for example, “1L milk” or “chocolate bar”), but it does not contain information specific to a particular unit of the product.

Expiration dates are usually determined by the **production batch or lot number**, rather than the general product type. Therefore, the expiration date is typically **printed directly on the product packaging** rather than stored in the barcode itself.

Because of this limitation, scanning a product barcode alone cannot reliably retrieve the expiration date of a specific item purchased by the user.

Possible Technical Solution

To address the challenges described above, this project proposes an alternative technical solution that uses **receipt image recognition and data analysis techniques** to extract purchase information and estimate food expiration dates.

1. Receipt Capture Using Mobile Camera

Instead of relying on POS system integration, the proposed system allows users to **capture a photo of their grocery receipt using the mobile application camera** after completing a purchase.

The captured receipt image will be uploaded to the system for further processing. This approach removes the dependency on retailer infrastructure and allows the application to work with receipts from **any supermarket or retail store**.

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2. Optical Character Recognition (OCR) for Text Extraction

Once the receipt image is captured, the system will apply **Optical Character Recognition (OCR)** technology to convert the receipt image into machine-readable text.

OCR techniques enable the system to detect and extract textual information from images, including:

- product names
- prices
- purchase date
- store information
- total spending amount

This step transforms the receipt image into raw text data that can be processed further by the system.

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3. Natural Language Processing for Receipt Analysis

After the receipt text is extracted using OCR, **Natural Language Processing (NLP)** techniques will be applied to analyze the text and identify relevant purchase information.

The NLP processing stage will perform tasks such as:

- identifying product names
- extracting price values
- grouping items within the receipt
- recognizing purchase dates and totals

The processed information will then be structured into a database format so that each purchased item can be stored as an individual record in the user's purchase history.

This structured data allows the system to support features such as spending analysis, product tracking, and purchase history queries.

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4. Food Category Classification

To support expiration prediction and spending analysis, the system will classify each product into a **food category** (for example, dairy, vegetables, snacks, beverages, etc.).

This classification can be achieved by using a **product database or keyword-based classification methods**. For example, if a product name contains the word "milk," it can be classified as a dairy product.

Categorizing products enables the system to associate each item with typical shelf-life information.

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5. Food Expiration Prediction Using Shelf-Life Database

Since expiration dates are usually not available directly from barcodes or receipts, the system will estimate the expiration date using a **food shelf-life database**.

This database will store typical shelf-life durations for different food categories. For example:

Food Category	Typical Shelf Life
Milk	7 days
Bread	3–5 days
Fresh vegetables	5–7 days
Yogurt	14–21 days

Food Category	Typical Shelf Life
Packaged snacks	Several months

When a product is recorded in the system, the application will combine:

Purchase date + shelf-life duration

to estimate the expected expiration date.

For example:

Purchase date: March 1
Product category: Milk
Estimated shelf life: 7 days

Estimated expiration date: March 8

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6. Expiration Monitoring and Notification

Once expiration dates are estimated, the system will continuously monitor the user's stored food items.

When a product approaches its predicted expiration date, the application will send **notifications or reminders** to the user. These reminders encourage users to consume food before it expires and help reduce unnecessary food waste.

Summary of the Proposed Solution

The proposed system replaces difficult POS integration with a **receipt image recognition pipeline**, consisting of:

1. Receipt image capture using mobile camera
2. OCR for text extraction
3. NLP for receipt information analysis
4. Product classification into food categories
5. Expiration prediction using a shelf-life database
6. Automated notification system for expiration reminders

This approach allows the application to function independently of retailer systems while still providing users with automated purchase tracking and food management capabilities.

The Problems still need to deal with

1. **Complexity of Automatically Extracting Receipt Information**

Even if digital receipts are obtained through other methods, extracting meaningful information from receipts can be challenging. Grocery receipts may have different formats depending on the store, and the text layout can vary significantly.

For example, receipts may contain:

- abbreviated product names
- irregular spacing
- different pricing formats
- promotional discounts

These variations make it difficult to automatically identify the product names, prices, and quantities using simple text processing techniques. Therefore, more advanced techniques are required to accurately extract structured information from receipt images.

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2. **Determining Product Expiration Dates**

Another challenge is determining the expiration date of purchased products when this information is not directly available from receipts or barcodes.

Different types of food have different shelf lives. For example:

- Milk may expire within several days.
- Bread may last only a few days.
- Packaged snacks may last several months.

Therefore, the system must have a method to **estimate expiration dates based on product categories and typical shelf-life data**, which requires building and maintaining a food expiration knowledge database.

System Architecture (Diagram-Level Explanation)

The proposed system will follow a **multi-layer architecture** consisting of a mobile application, backend processing services, and a database system. This architecture allows the system to efficiently capture receipt data, process information using machine learning techniques, and provide intelligent feedback to users.

At a high level, the system consists of four main components:

1. **Mobile Application Layer**
2. **Image Processing and Data Extraction Layer**
3. **Backend Processing and Data Analysis Layer**
4. **Database and Knowledge Base Layer**

These components work together to capture receipt images, extract purchase information, store the data, and generate insights for the user.

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1. Mobile Application Layer

The mobile application serves as the **primary user interface** of the system. Users interact with the system through this application to capture receipts, view purchase history, and receive food expiration reminders.

The mobile application provides several key functions:

- Capturing receipt images using the smartphone camera
- Uploading receipt images to the backend server
- Displaying extracted purchase information
- Showing spending analysis and purchase history
- Providing food expiration notifications
- Offering AI-based shopping suggestions

The mobile application sends receipt images to the backend server for processing and receives structured purchase data in return.

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2. Image Processing and Data Extraction Layer

After a receipt image is uploaded, the system processes the image using **Optical Character Recognition (OCR)** techniques.

This layer performs the following tasks:

1. **Image preprocessing**
   - Improve image quality
   - Adjust brightness and contrast
   - Detect receipt boundaries
2. **Text extraction using OCR**
   - Detect text regions in the receipt image
   - Convert image text into machine-readable text

The extracted raw text will include information such as product names, prices, purchase date, and store information.

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3. Backend Processing and Data Analysis Layer

Once the receipt text has been extracted, the backend system performs **Natural Language Processing (NLP)** and data analysis to structure the receipt data.

The backend processing layer performs several operations:

- Identifying individual purchased items from the receipt text
- Extracting product names, prices, and quantities
- Organizing items into structured purchase records
- Classifying products into predefined food categories

This structured data is then stored in the database and used for additional analysis.

The backend also includes an **AI-based recommendation module** that analyzes user purchasing behavior and provides personalized shopping suggestions.

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4. Database and Knowledge Base Layer

The database layer stores all relevant information required by the system. This includes:

- User account information
- Purchase history records
- Product categories
- Food shelf-life data
- Predicted expiration dates

The **food shelf-life database** plays an important role in the system. It stores typical shelf-life durations for different types of food products. When a purchase is recorded, the system uses this information to estimate the expiration date of each item.

For example:

Estimated expiration date = Purchase date + Typical shelf life of the product category

This estimated expiration date is stored in the database and monitored by the system.

System Data Flow

The overall system workflow can be summarized as follows:

1. The user captures a receipt image using the mobile application.
2. The image is uploaded to the backend server.
3. OCR technology extracts text from the receipt image.
4. NLP techniques analyze the extracted text and identify purchase items.
5. Each item is categorized using the product classification module.
6. The system estimates expiration dates using the food shelf-life database.
7. All structured data is stored in the system database.
8. The mobile application displays purchase history, spending insights, and expiration reminders to the user.

This architecture enables the system to automatically transform unstructured receipt images into structured purchase records and provide intelligent insights for users.

(If you draw a diagram in your report, it will typically look like this flow):

Mobile App → Receipt Image → OCR Engine → NLP Processing → Product Classification → Database → AI Recommendation → User Interface

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Expected Outcomes and Evaluation Methods

The primary goal of this project is to develop a working prototype of a **Smart Digital Receipt and Food Expiration Management System** that can automatically extract purchase information from receipt images and assist users in managing their grocery consumption.

The success of the project will be evaluated based on several technical and usability outcomes.

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1. Accurate Extraction of Receipt Information (ORC)

One expected outcome is the ability of the system to accurately extract purchase information from receipt images.

The system should correctly identify key information such as:

- Product names
- Prices
- Purchase date
- Total spending amount

To evaluate this component, a dataset of grocery receipt images will be collected and processed by the system. The extracted information will then be compared with manually labeled ground truth data.

Evaluation metrics may include:

- **Extraction accuracy**
- **Item recognition rate**
- **Error rate in price detection**

This evaluation will demonstrate how effectively the OCR and NLP components perform in real-world conditions.

2. Effectiveness of Product Classification

Another expected outcome is the ability of the system to classify purchased items into appropriate food categories.

For example:

Milk → Dairy
Apple → Fruit
Bread → Bakery

The accuracy of this classification process can be evaluated by comparing the system's predictions with manually assigned categories.

Evaluation metrics may include:

- Classification accuracy
- Precision and recall for each category

This evaluation helps determine whether the system can reliably organize purchase data for further analysis.

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3. Accuracy of Expiration Date Prediction (Machine Learning -> NLP)

Since expiration dates are estimated based on a shelf-life database, the system must provide reasonably accurate expiration predictions.

To evaluate this feature, a set of food items with known expiration dates will be used as reference data. The system's predicted expiration dates will be compared with the actual expiration dates printed on the product packaging.

Evaluation metrics may include:

- Average prediction error (in days)
- Percentage of predictions within an acceptable time range

This evaluation measures how effectively the system can estimate expiration dates for different types of food.

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4. User Experience and System Usability

Another important outcome is the usability of the mobile application.

The system should allow users to easily:

- capture receipt images
- view purchase history
- receive expiration reminders
- review spending analysis

User testing can be conducted with a small group of participants who will use the prototype application and provide feedback regarding usability and convenience.

Evaluation methods may include:

- user satisfaction surveys
- task completion time
- qualitative user feedback

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5. Overall System Performance

The final outcome of the project is a fully integrated prototype that demonstrates the feasibility of the proposed system.

The system should successfully perform the following tasks:

- capture receipt images
- extract purchase information automatically
- classify products into categories
- estimate expiration dates
- notify users of upcoming food expiration
- generate shopping insights using historical purchase data

Demonstrating the successful integration of these components will show that the proposed concept can be implemented using current technologies.

5W1H Analysis of the Project

Who

The primary users of this system are **grocery shoppers and households** who regularly purchase food from supermarkets. These users often need better tools to track their grocery spending and manage food consumption at home. In addition, people who want to reduce food waste and improve personal budgeting can benefit from this application.

What

This project proposes a **Smart Digital Receipt and Food Expiration Management Application**. The system captures grocery receipts using a mobile camera, automatically extracts purchase information, stores digital receipts, tracks spending behaviour, predicts food expiration dates, and provides intelligent shopping suggestions using AI analysis.

When

This system can be used **immediately after grocery shopping**, when users capture their receipt using the mobile application. The stored information can then be accessed anytime to review purchase history, monitor spending patterns, and receive notifications when food items are approaching their expiration dates.

Where

The system will primarily operate on **mobile devices such as smartphones**, allowing users to access their purchase records and food management tools anywhere. The backend processing and data storage will be handled through a cloud-based server and database system.

Why

Many people receive paper receipts when shopping but rarely keep or organize them. As a result, it is difficult for consumers to track their grocery spending or remember what food they have purchased. Additionally, food waste is a common problem because people often forget about expiration dates of stored food. This project aims to solve these issues by providing an automated digital system that helps users track purchases, manage food expiration, and make smarter shopping decisions.

How

The system works by allowing users to **capture receipt images using their smartphone camera**. Optical Character Recognition (OCR) extracts text from the receipt, and Natural Language Processing (NLP) analyzes the purchase information. The system then classifies food items into categories and estimates expiration dates using a shelf-life database. Based on the stored purchase data, the application can provide spending insights, expiration reminders, and AI-based shopping recommendations.

My opinion:

I don't know why this project idea will come to my mind. Just one day, when I was lying on my sofa, the idea jumped into my head. So, I probably don't have any reference and any research talking about this topic. I just feel like it is convenient if someone can remind me what I have put into my refrigerator. All the above, are from the chat box with ChatGPT. I asked it, if my original idea feasible, and it told me a lot of problems that we might face if we choose my topic. (I would be really happy if we can choose another topic which is easier and more interesting) I don't want u guys to be stressful after reading this document, I put zero effort in this one. It seems scary, since it's more than 17 pages. But believe me, I just want to prove that I have done something XD. (I want to be a good teammate). Btw, Tammy told me that the function 1 and function 2 above, have quite similar apps. For function 1, in Taiwan, people will show their mobile barcode to the staff to get the receipt. These receipts are the evidence that the store has paid the consumer tax to the government. And the number on the receipt, people have the chance to win the lottery. For functional 2, like in Commonwealth bank, it categorizes for u after u paid by credit card. These two functions are not innovation, which is really important in this subject.